

Simplified numerical modeling with experimental validation of molten pool geometry in laser welding

Mahdi Ghamari^a, Pouria Oliazadeh^b, Mohammad Hossein Farshidianfar^{b,*}

^a Department of Mechanical Engineering, Sharif University of Technology, Tehran, , Iran

^b Department of Mechanical Engineering, Ferdowsi University of Mashhad, Mashhad, Iran

ARTICLE INFO

Keywords:

Laser welding
Numerical simulation
Molten pool geometry
Phase-change modeling
Gaussian heat source

ABSTRACT

Laser welding is widely adopted for its high speed, precision, and automation capabilities, delivering superior microstructural properties in material joining. However, suboptimal welding parameters can cause defects, necessitating accurate and computationally efficient simulation models. This study introduces a novel simplified numerical model for laser welding that optimizes the prediction of molten pool geometry while minimizing computational cost. By integrating a phase-change model with a calibrated Gaussian heat source, the approach balances accuracy and efficiency, addressing the limitations of complex fluid dynamics models and over-simplified temperature-based simulations. The model was validated against seven experimental cases involving Ti6Al4V and stainless steel 316 L, achieving zero error in melt pool depth prediction and maximum errors in top and bottom widths of <0.2 mm. This computationally efficient model enables rapid parameter optimization, making it ideal for large-scale industrial applications, including residual stress and distortion analysis. By offering a practical alternative to high-cost simulations, this work advances laser welding simulation for precision manufacturing and supports broader adoption in industries such as automotive and railway.

1. Introduction

Welding remains a crucial technique for joining metals, alloys, and even plastics by applying heat and pressure at the interface. With the evolution of welding technologies, over 75 methods are now available, many of which are combinations of various techniques [1]. These methods are typically categorized based on heat input, including Arc welding, Gas welding, Resistance welding, Solid-state welding, and High-intensity welding. Laser welding, classified under high-intensity welding, has garnered significant attention due to its advantages, such as high processing speeds, automation capabilities, fined microstructures, and enhanced strength. These attributes make it particularly favored in industries such as automotive, railway, and heat transfer [1].

As laser welding gains broader usage, the need for accurate simulation models grows. However, given the multitude of factors that influence the weld outcome, identifying optimal welding parameters through traditional trial-and-error methods is both costly and time-consuming. Key parameters, such as welding speed, focal length, input power, and material type, significantly impact the final weld quality. Improperly tuned parameters can result in defects like distortion, incomplete joints, and residual stresses.

A critical aspect that determines weld quality is the shape of the molten zone [2,3]. Poorly shaped melt pools can lead to stress concentration points, weakening the welded joint. Given the strong relationship between the melt zone geometry and weld performance, accurate simulation models are crucial in preventing defects like porosity, distortion, cracks, and stress concentration. Since welding involves phase changes, which depend on temperature, achieving accurate temperature distribution in simulations is essential for precise weld predictions.

The first significant advancement in welding simulation came in 1941, when Rosenthal [4] applied Fourier's law and a moving point heat source to model weld behavior. However, this approach concentrated all energy into a point, failing to predict the melted zone accurately. Pavelic later proposed a line heat source, addressing some limitations but still lacking accuracy in temperature distribution [5]. Over time, volumetric heat sources, introduced by Friedmann [6] and later expanded by others, became the standard for predicting the temperature field and molten pool shape. In 1983, Eagar and Tsai [7] developed a non-uniform volumetric heat source, which became a foundation for modern heat source models. These models have been further refined, with researchers developing conical, cylindrical, and cubic heat sources to better simulate

* Corresponding author at: Mechanical Engineering, Ferdowsi University of Mashhad, Azadi Square, Mashhad, Razavi Khorasan Province, Iran.

E-mail addresses: mahdi_ghamari@mech.sharif.edu (M. Ghamari), pouria_oliazadeh@mail.um.ac.ir (P. Oliazadeh), farshidianfar@um.ac.ir (M.H. Farshidianfar).

molten pool geometry [8,9], keyholes [10–15], and even porosity formation [13,16]. More recently, Stavropoulos et al. [17] introduced a fast-running model that incorporates laser defocusing and effective reflectivity in keyhole processes, representing an advancement in capturing keyhole-related phenomena in numerical models.

As heat source models evolved, researchers also began investigating residual stresses and distortion in welded joints. Early work in this field by Rodger and Fletcher [18] (1938) did not yield significant results due to computational limitations. The first practical simulation using the finite element method (FEM) came from Brust et al. [19–21] in 1970, and since then, FEM has been widely applied to study weld profiles, temperature gradients, heat source calibration, and residual stresses. Sattari-Far et al. [22] explored the effect of multiple welding passes on distortion and residual stress in steel tubes, concluding that increasing pass numbers did not significantly mitigate welding side effects.

Abedin et al. [23] examined the influence of single-pass welding on residual stress and demonstrated the critical role of heat source selection in determining the final weld zone shape. Further studies by Kim et al. [24] investigated different heat sources in a T-joint and then developed a special heat source for this type of joint. Chukkan et al. [25] evaluated various heat sources, demonstrating that while heat source type does not drastically affect peak temperatures, it plays a crucial role in determining weld zone geometry, residual stress, and distortion. Casalino and et al. [26] emphasized the importance of calibrating heat sources with experimental data for accurate laser welding simulations.

With advances in fluid mechanics, researchers began incorporating fluid dynamics into laser welding simulations. These models allowed for the exploration of previously inaccessible aspects of welding, such as phase changes, flow regimes, and keyhole dynamics. Unlike FEM-based models, which focused solely on temperature distribution, fluid dynamics simulations provided insights into the behavior of the molten pool, keyhole instability, and porosity formation.

Feng et al. [27] investigated the molten pool and keyhole dynamics in different penetrations during the welding process, while Zhou and et al. [28] developed a two-dimensional heat source model to simulate porosity formation. However, these studies often overlooked the impact of welding speed on simulation accuracy [29]. To improve accuracy, Cho et al. [30] proposed a three-dimensional heat source model for hybrid welding, capturing keyhole instability and bubble formation in the molten pool. Zhao and et al. [31] employed ANSYS Fluent to model keyhole dynamics, identifying keyhole collapse as the primary mechanism behind porosity formation. Also, Coleman et al. [32] proposed a generalized two-parameter volumetric heat source model that integrates existing approaches into a unified formulation, allowing improved calibration of melt pool geometry across conduction and keyhole regimes.

As previously mentioned, laser welding simulations can generally be categorized into two main approaches: (1) temperature distribution analysis without considering phase change and (2) keyhole dynamics analysis, including the formation of porosity within the weld region. A primary limitation of models that account solely for temperature distribution, while overlooking phase change, is the uncertainty in predicting the molten pool profile and its associated shape under varying conditions [33]. On the other hand, models that incorporate phase change, while capable of accurately predicting the molten pool shape and porosity formation, tend to be highly complex and computationally expensive. While previous models have focused on capturing keyhole dynamics and porosity formation, their complexity and computational expense limit their practical application in large-scale welding simulations [8,34]. Despite these advancements, there remains a trade-off between accuracy and computational cost in laser welding simulations.

In this study, a simplified phase-change model is proposed to accurately predict molten pool geometry and temperature distribution while substantially reducing computational costs. Complementary to our approach, Giudice and Sili [33] validated a simplified analytical thermal-field model against multiphysics CFD simulations,

demonstrating the ongoing relevance of reduced-order methods. Additionally, a recent review by Jiménez-Xamán et al. [35] categorized modern laser welding heat-source models, further highlighting the research gap that motivates the present work. In contrast to previous models that either oversimplify the phase-change process or entail excessive computational complexity, the proposed approach achieves a balanced compromise. It effectively captures the key thermal and geometric characteristics of the molten pool with reduced computational demand, rendering it a practical solution for large-scale industrial applications, particularly in the analysis of residual stresses and distortions.

2. Numerical modeling

2.1. Governing equation

Welding is an inherently complex process involving phase changes, mass transfer, heat transfer, and fluid flow. Simulating all of these phenomena simultaneously can be computationally expensive or even impractical. To address this, a simplified model can be employed, focusing on the most critical aspects while neglecting phenomena of lesser importance, yet still delivering results within an acceptable range of accuracy.

In this simplified model, fluid flow is assumed to be laminar and incompressible, despite the reality that welding often involves turbulent flow. Furthermore, while all three types of heat transfer—conduction, convection, and radiation—are present in welding, radiation is omitted from this model. This omission is justified because radiation is proportional to the fourth power of temperature and surface area, and only a small portion of the welding domain reaches temperatures high enough for radiation to have a significant effect.

Material properties such as thermal conductivity, heat capacity, density, latent heat, and viscosity are considered temperature-dependent, with viscosity assumed to be Newtonian in the liquid phase. These properties were extracted from experimental results and from JMatPro software, as reported in references [36–40]. Although the welding process includes both solid-to-liquid and liquid-to-vapor phase changes, the latter is neglected for the sake of simplifying the model without compromising overall accuracy.

2.1.1. Mass conservation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = S_m \quad (1)$$

Eq. (1) represents the general form of the mass conservation equation, applicable to any fluid flow. In this equation, ρ denotes the density, which is a function of temperature, $\vec{V}$ is the velocity vector, and S_m represents the source term for mass in the continuous phase.

2.1.2. Momentum equation

$$\frac{\partial (\rho \vec{V})}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = -\nabla P + \nabla \cdot \left[\mu (\nabla \vec{V} + \nabla \vec{V}^T) - \frac{2}{3} \mu \nabla \cdot \vec{V} I \right] - S + \rho \beta \vec{g} (T - T_m) \quad (2)$$

$$S = \frac{(1 - \beta)^2}{(\beta^3 + \varepsilon)} A_{mush} (\vec{V} - \vec{V}_p) \quad (3)$$

Eq. (2) is the momentum equation, with two additional terms. The first, S , is used in the mushy zone, where the enthalpy-porosity model is applied [41]. In this model, the mushy zone is treated as a porous medium, with porosity being a function of the liquid fraction. To solve the fluid flow equation in this region, the term S must be included in the momentum equation.

The last term on the right-hand side of Eq. (2) represents the

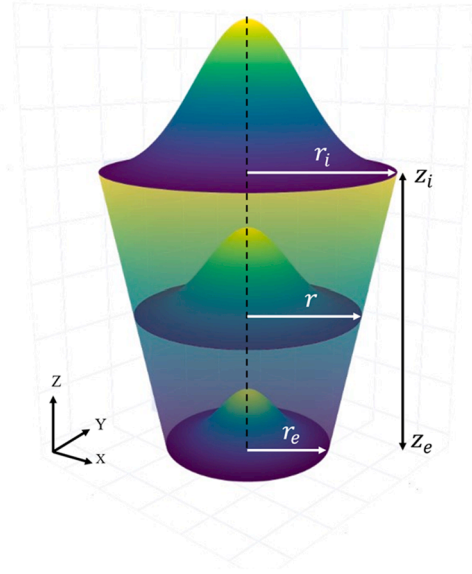


Fig. 1. Truncated conical Gaussian heat source model used in the simulation.

buoyancy force, derived from the Boussinesq approximation [42]. Other variables include P , representing pressure; μ , viscosity; β , the thermal expansion coefficient; and T_m , the melting temperature.

2.1.3. Energy equation

$$\frac{\partial(\rho H)}{\partial t} + \nabla \cdot (\rho \vec{V} H) = \nabla \cdot (k \nabla T) + S_E \quad (4)$$

In Eq. (4), which represents the energy equation, H denotes enthalpy, k is the thermal conductivity, and S_E is the energy source term. The enthalpy of the material is calculated as the sum of the sensible enthalpy, h , and the latent heat, ΔH .

$$H = h + \Delta H \quad (5)$$

where:

$$h = h_{ref} + \int_{T_{ref}}^T c_p dT \quad (6)$$

h_{ref} = reference enthalpy

T_{ref} = reference temperature

c_p = specific heat at constant pressure

Latent heat is associated with phase changes, so this term must account for the amount of each phase present. The variable β , known as the liquid fraction, is used for this purpose and is defined as:

$$\beta = 0 \text{ if } T < T_{solidus} \quad (7)$$

$$\beta = 1 \text{ if } T > T_{liquidus} \quad (8)$$

$$\beta = \frac{T - T_{solidus}}{T_{liquidus} - T_{solidus}} \text{ if } T_{solidus} < T < T_{liquidus} \quad (9)$$

Thus, the latent heat can be expressed as a function of the liquid fraction, β , and the material's latent heat, L :

$$\Delta H = \beta L \quad (10)$$

2.2. Laser heat source

A truncated heat source with a Gaussian energy distribution is employed in the laser welding simulation [25,43,44]. The shape of the heat source is presented in Fig. 1, and it can be expressed as:

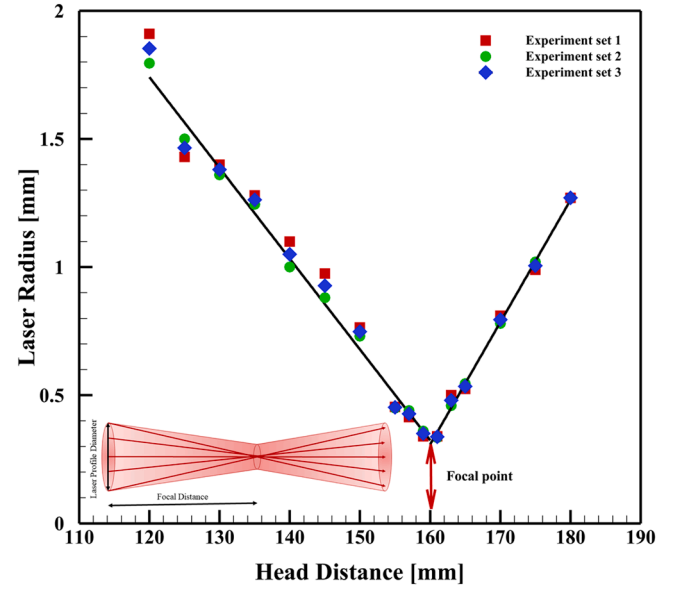


Fig. 2. Radius variational of laser profile in different head-distance.

$$Q(x, y, z) = Q_0 \exp\left(-\frac{x(t)^2 + y(t)^2}{r_0(z)^2}\right) \quad (11)$$

$$r_0(z) = r_e + \frac{r_i - r_e}{z_i - z_e}(z - z_e) \quad (12)$$

$$x(t) = vt + x_0 \quad (13)$$

$$y(t) = vt + y_0 \quad (14)$$

The position variables, $x(t)$ and $y(t)$, correspond to the axis of the heat source, with the Gaussian function representing the energy distribution around this axis. Q_0 denotes the power density, defined as the laser power divided by the volume of the heat source. r_i and r_e represent the heat source radius, which are assumed to be equal to the laser profile radius.

To calibrate the variation in the laser beam profile diameter as a function of head distance, an experimental procedure was conducted using a leather sample. The leather was irradiated at maximum laser power at varying head distances, and the dimensions of the affected area were measured. Fig. 2 illustrates the relationship between the laser beam profile radius and head distance.

To ensure the reliability and reproducibility of the experimental results, measurements were performed three times under identical conditions. The average values obtained were used to derive a mathematical function describing the dependence of the laser beam profile diameter on head distance. This function was subsequently utilized to estimate the upper and lower diameters of the heat source profile in further simulations and analyses.

2.3. Boundary condition

In the laser welding model, the initial domain temperature is set to room temperature, and only the metal domain is considered in the simulation. However, in reality, some of the liquid metal evaporates, generating recoil pressure in the keyhole. This recoil pressure pushes the surrounding liquid metal aside, resulting in a wider molten pool. Since evaporation is neglected in the model for simplicity, a substitute is needed to replicate the effect of recoil pressure.

The solution is the implementation of an artificial wall. By placing an artificial wall at the top and bottom of the liquid domain, the liquid metal is pushed aside, simulating the behavior of recoil pressure. The

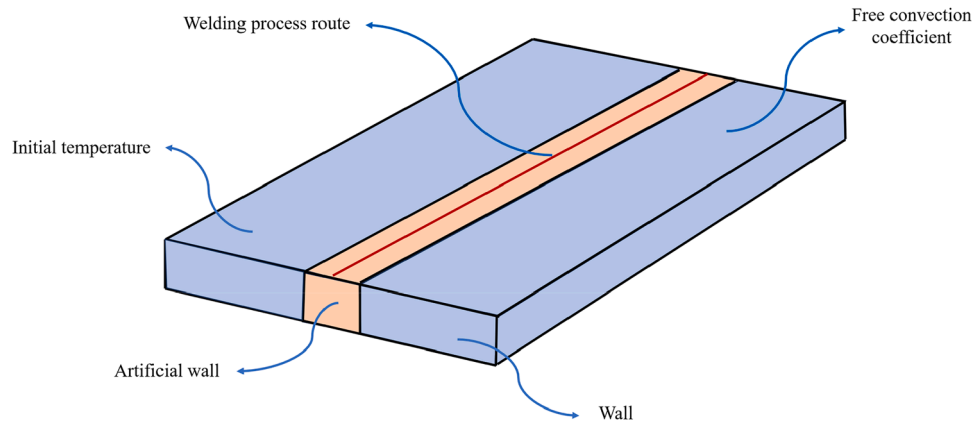


Fig. 3. Boundary condition schematic for laser welding simulation, including artificial wall used to replicate recoil pressure effects.

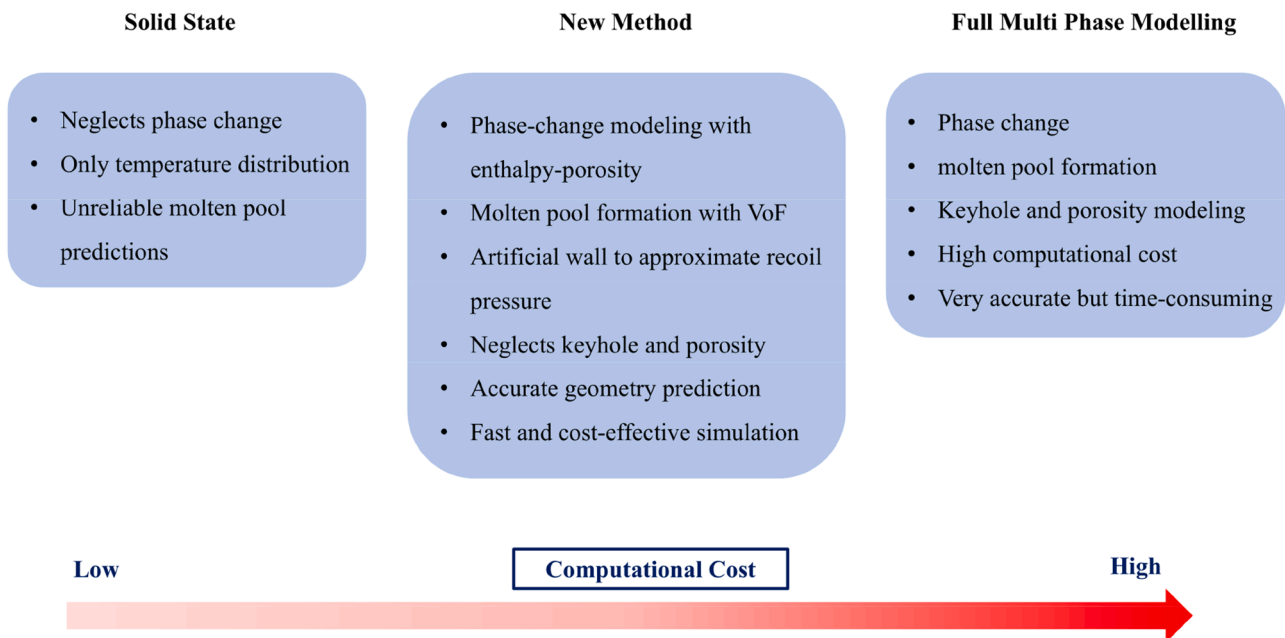


Fig. 4. Comparison of solid-state, proposed simplified, and full multiphase laser welding models in terms of physics captured and computational cost.

domain wall is assumed to undergo free convection as if on a horizontal

plate, with heat transfer occurring between the plate and the environment. A schematic of the boundary conditions is provided in Fig. 3.

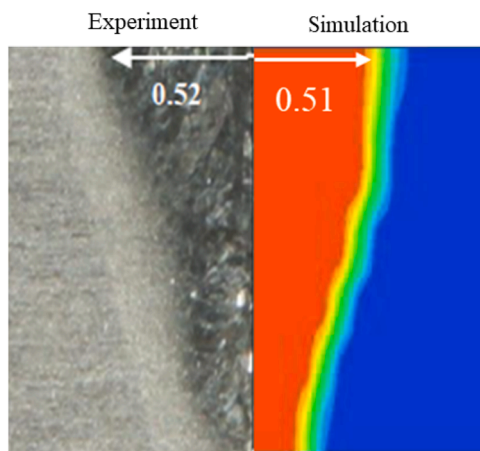


Fig. 5. Validation of the numerical model against experimental molten pool geometry (units in mm).

2.4. Simplified modeling methodology

The proposed methodology offers a balanced approach between purely thermal solid-state models and computationally intensive full multiphase simulations. Unlike solid-state models, which neglect phase change and therefore provide uncertain predictions of molten pool geometry, the simplified model explicitly accounts for phase-change phenomena using the enthalpy-porosity formulation and resolves molten pool formation with the Volume of Fluid (VoF) method. Keyhole dynamics and porosity formation are neglected, but their effects are approximated by introducing an artificial wall to mimic recoil pressure. This strategy significantly reduces computational cost while maintaining high accuracy in predicting molten pool depth and acceptable precision in pool width. As illustrated in Fig. 4, the proposed model therefore combines the accuracy advantages of multiphase simulations with the efficiency of simplified approaches, making it particularly suitable for rapid parameter studies and industrial applications.

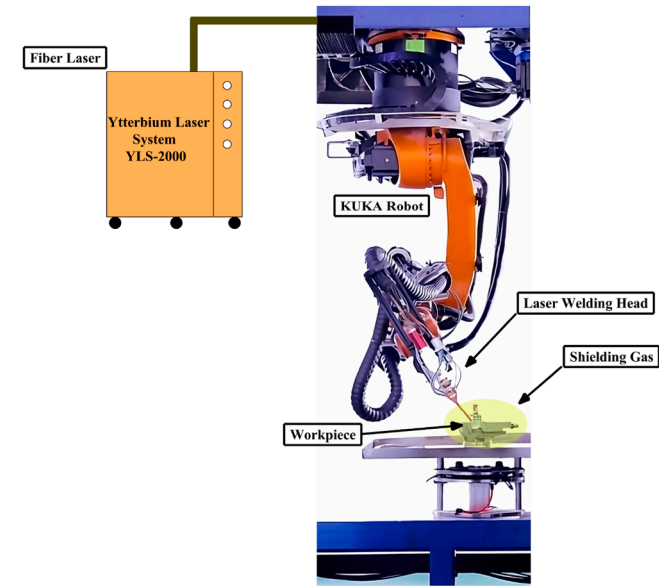


Fig. 6. Schematic of the experimental setup, showing robot-assisted laser head movement and Argon shielding.

3. Simulation

3.1. Model validation

The experimental parameters from reference [26] have been used for model validation, and the experiment outline is described here. In the experiment, two rectangular pieces of Ti6Al4V titanium alloy were welded using the butt weld method. Each piece had dimensions of 1.5 mm in thickness, 10 mm in width, and 20 mm in length. A Ytterbium Laser System with a maximum power output of 4 kW was utilized. The laser beam had a wavelength of 1076.6 nm, was focused using a lens with a focal distance of 250 mm, and had a spot diameter of 0.4 mm.

In the reference case, the laser power was set to 1.5 kW, with a welding speed of 3500 mm/min and a focus diameter of 300 μ m. Using these experimental parameters, the previously explained model was simulated. The simulation results were excellent. The depth of the melted zone in the simulation matched the experimental result perfectly, and the top width showed only a 1.92 % error compared to the experiment. A comparison of the simulation and experimental results is shown in Fig. 5.

3.2. Experimental setup

In this study, the experimental setup features a Ytterbium Laser System with a maximum output power of 2 kW. Welding motion is achieved using a Kuka robot arm while the workpieces remain stationary. The robot arm moves along the welding path and adjusts the laser head to the appropriate height above the surface. Argon shielding gas is used during welding to prevent oxidation of the molten pool. A schematic of the experimental setup is provided in Fig. 6.

Table 1
Experiment cases parameters.

Case	1	2	3	4	5	6
Material	Stainless steel 316L	Stainless steel 316L	Stainless steel 316L	Stainless steel 316L	Stainless steel 316L	Ti6Al4V
Thickness (mm)	2	2	1	1	2	2.7
Speed (mm/s)	38	13.8	40	14	25	15
Power (W)	1400	1400	1950	1950	1900	1900
Head distance (mm)	160	160	170	170	155	160

3.3. Experiment case simulation

Six simulation cases were conducted. Five of the cases involved stainless steel 316 L, while the remaining case used Ti6Al4V. The parameters for each experiment are presented in Table 1. All experiments were carried out at room temperature. The first step in the simulation

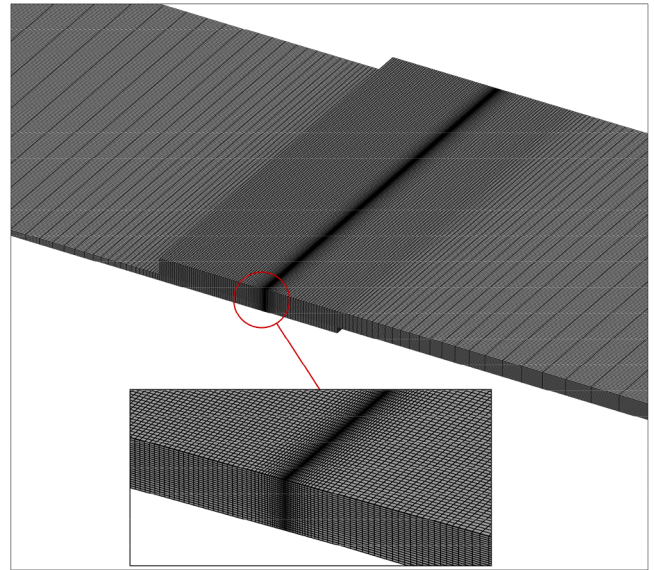


Fig. 7. Mesh generation for the numerical domain, emphasizing high-resolution near the weld zone.

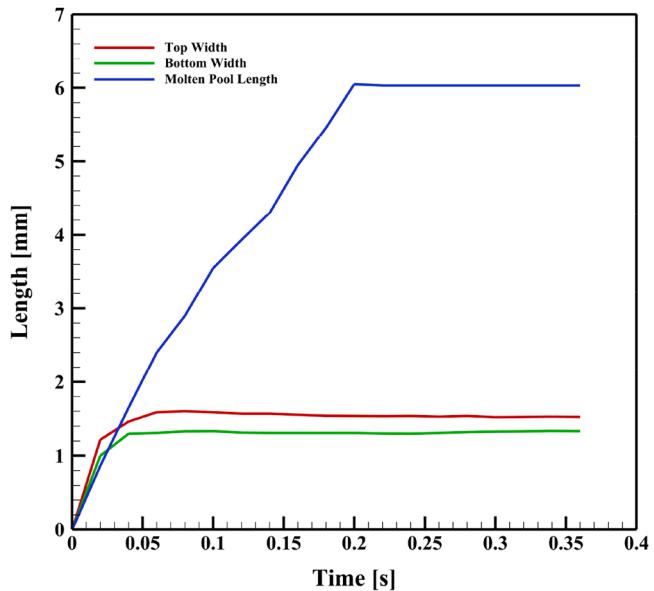


Fig. 8. Evolution of molten pool dimensions over time for experimental case 1, showing stabilization at steady state.

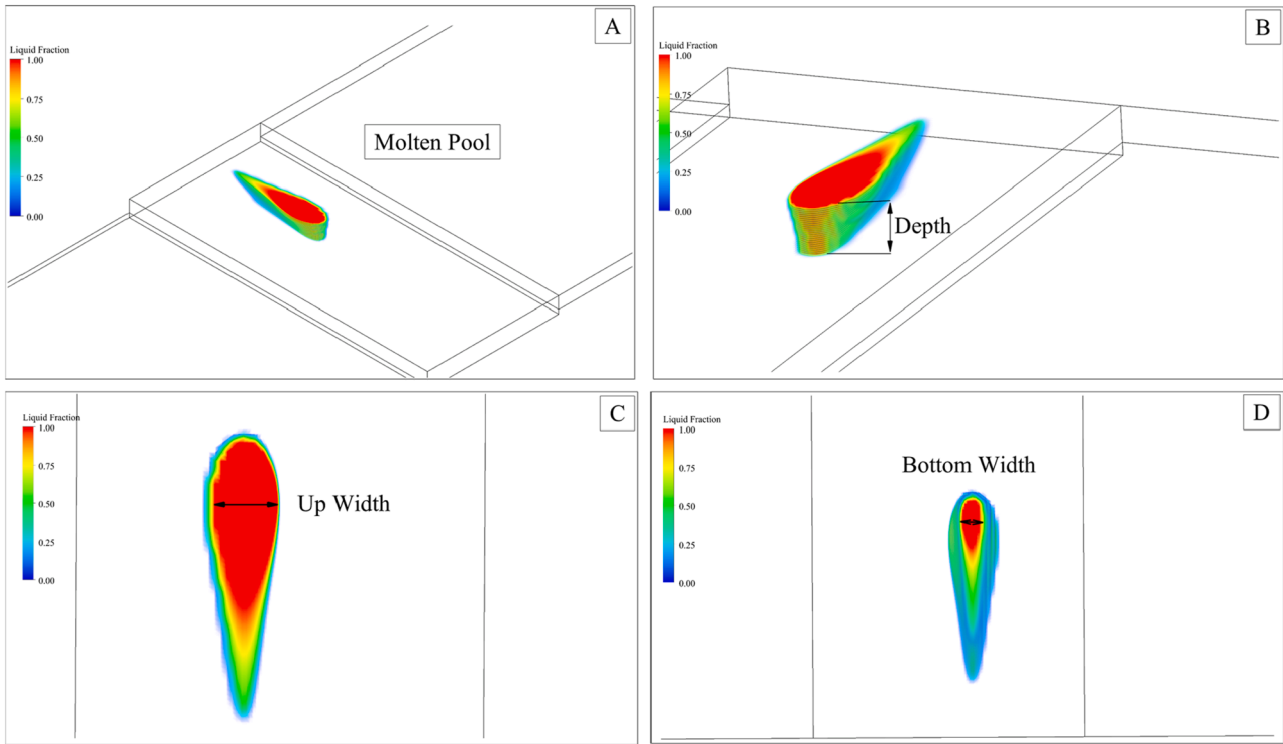


Fig. 9. (A) Simulation overview; (B) Melt pool depth; (C) Top width; (D) Bottom width of the molten pool.

process involved discretizing the geometry. A sample of the mesh generation is shown in Fig. 7. Mesh and time step independence are critical for achieving accurate results, and after multiple refinements, the optimal parameters were determined. The mesh near the welding path had a width of 0.05 mm with a growth rate of 1.05, and the best time step was set at 0.02 s.

It is important to note that the simulation was performed using finite volume method software, with the Volume of Fluid (VoF) model applied to account for multiphase flow. Material properties such as density, thermal conductivity, and specific heat varied with temperature during the simulations. A regression was performed on the material properties data, and the corresponding functions were implemented in the simulation software.

The heat source was defined based on a truncated Gaussian energy distribution, with parameters such as r_l , r_e , and volume heat source adjusted relative to the head distance. Since the laser welding process is inherently dynamic rather than a static snapshot, it is necessary to measure the molten pool dimensions after the system reaches a steady state. As illustrated in Fig. 8, the variations in molten pool geometry stabilize over time, and this point is taken as the reference for comparing the simulation results with the experimental observations. After the simulation was completed, the width and depth of the melted zone were measured, as shown in Fig. 9. The results of the simulation were then compared with the experimental outcomes.

3.4. Computational efficiency

All simulations in this study were carried out on a personal workstation equipped with an Intel Core i7 processor (4 cores) and 12 GB RAM. A representative case (stainless steel 316 L, Case 1) required approximately 40 min of wall-clock time to simulate a physical process of 0.35 s. Keyhole-resolved CFD models require substantially higher computational cost because the vapor region must also be resolved: all governing equations (mass, momentum, and energy) must be solved for the vapor phase in addition to liquid and solid phases, and an additional phase-equilibrium equation must be solved simultaneously. Moreover,

keyhole modeling requires much finer meshes to capture recoil pressure, vaporization, and multiphase flow. These factors significantly increase runtime and memory demands; in contrast, the proposed reduced-order model produces accurate molten-pool predictions in under an hour on a standard desktop, making it suitable for rapid parameter optimization and industrial prototyping.

4. Results and discussion

The simplified numerical model for laser welding was validated against seven experimental cases, including a reference study [26] and six additional experiments involving Ti6Al4V and stainless steel 316 L. The model's predictive accuracy was evaluated by comparing simulated molten pool geometries (depth, top width, bottom width) with experimental measurements, as presented in Fig. 10. Additionally, the molten pool length, a critical parameter influencing weld quality and process stability, was analyzed based on simulation results, as shown in Fig. 11. This section provides a comprehensive analysis of the model's performance, the influence of welding parameters on molten pool geometry and length, and the implications for industrial applications. The discussion also addresses the model's limitations and its computational efficiency compared to more complex approaches.

4.1. Predictive accuracy of molten pool geometry

The model accurately predicts the depth of the molten pool across all experimental cases, achieving zero error relative to experimental measurements (Fig. 10a). This precision is attributed to the calibrated truncated Gaussian heat source, which effectively models the laser energy distribution, and the enthalpy-porosity model, which captures phase transitions in the mushy zone. The top and bottom widths of the molten pool exhibited maximum errors of 24.55 % and 19.71 %, respectively, with minimum errors of 0.00 % and 3.57 % (Fig. 10b and c). These errors, corresponding to absolute deviations of <0.2 mm, are within acceptable limits for industrial applications, particularly given the model's simplified assumptions, such as the neglect of keyhole

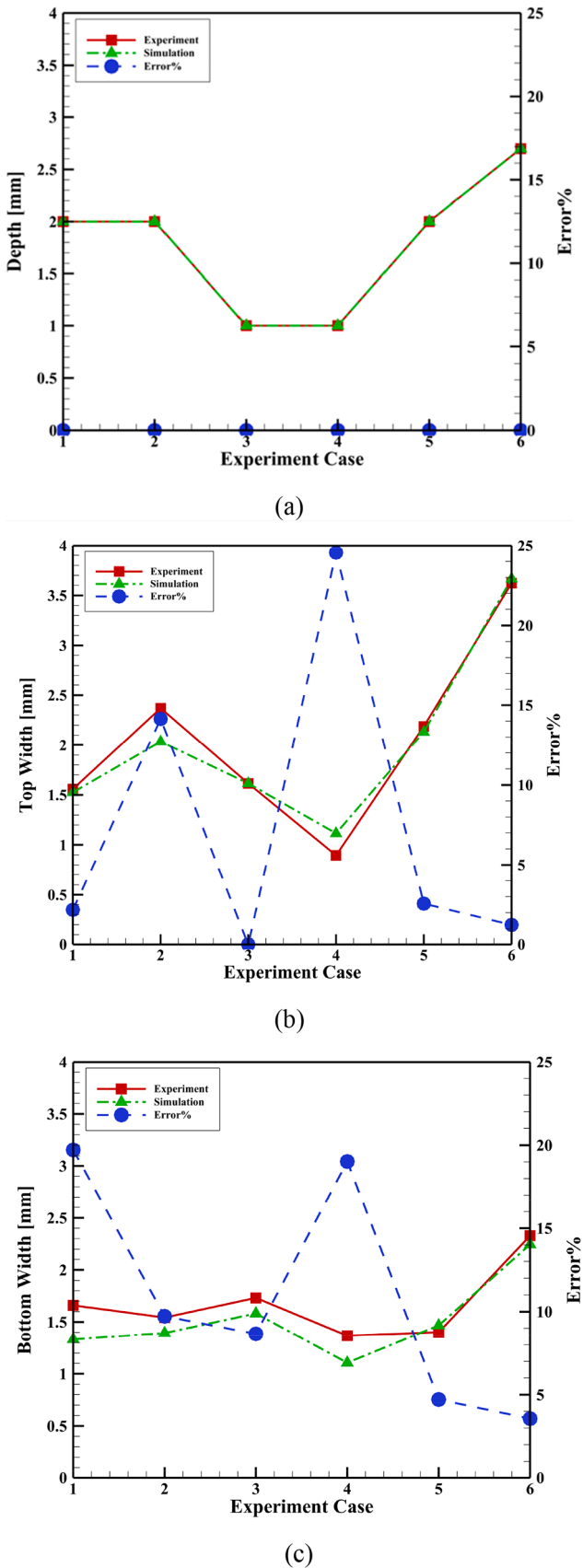


Fig. 10. Comparison between experimental and simulated results for molten pool geometry: (a) depth, (b) top width, and (c) bottom width.

dynamics and metal vaporization.

The high accuracy in depth prediction highlights the model's ability to capture the thermal energy required for melting, while errors in width predictions arise from the omission of complex fluid dynamics and recoil pressure effects. For instance, in the reference case [26], the model predicted the top width with a 1.92 % error, demonstrating its capability under well-calibrated conditions. However, larger errors in cases such as case 4 (top width error ~ 24.55 %) indicate that the simplified heat source model may not fully account for lateral molten pool spread in deep penetration welding scenarios where keyhole dynamics are significant.

4.1.1. Effect of welding speed on molten pool geometry

The influence of welding speed on molten pool geometry was investigated in cases 1 through 4. In cases 1 and 2, where the laser head distance was below the focal distance (160 mm), reducing the welding speed from 38 mm/s to 13.8 mm/s (case 2) increased the top width of the molten pool, while the bottom width remained nearly unchanged (Fig. 10b and c). This trend suggests that lower welding speeds increase heat input per unit length, promoting greater surface melting. The stability of the bottom width indicates that penetration depth is primarily governed by laser power and beam focus.

In contrast, cases 3 and 4, with a head distance of 170 mm (exceeding the focal distance), showed a different behavior. Reducing the welding speed from 40 mm/s (case 3) to 14 mm/s (case 4) decreased both top and bottom widths, with the bottom width exceeding the top width in case 4 (Fig. 10b and c). This inversion is attributed to multiple laser beam reflections within the keyhole, which enhance energy absorption at greater depths when welding speed is reduced, as illustrated in Fig. 12. The increased heat input at lower speeds facilitates deeper penetration and a wider molten pool at the bottom, emphasizing the critical role of head distance in determining weld profiles in deep penetration welding.

4.1.2. Influence of material and parameter variations

Cases 5 and 6 tested the model's robustness by varying material (stainless steel 316 L in case 5, Ti6Al4V in case 6) and parameters such as laser power and head distance. The model consistently predicted molten pool depth with zero error, while width predictions remained within acceptable margins (maximum error ~ 4.71 % for bottom width in case 5).

Errors in width predictions, particularly in case 4, stem from the model's simplified assumptions. Neglecting keyhole formation and metal vaporization limits the model's ability to capture complex fluid dynamics and recoil pressure effects, which influence the lateral spread of the molten pool. Despite these limitations, the model's computational efficiency and acceptable accuracy make it a practical tool for preliminary parameter optimization in industrial settings.

4.2. Analysis of molten pool length

Fig. 11 presents the simulated molten pool length, defined as the longitudinal extent of the molten zone along the welding direction, for the six experimental cases. The molten pool length is a critical indicator of heat input distribution, cooling rates, and process stability, directly impacting weld quality and defect formation. The simulated lengths provide insights into the model's ability to capture longitudinal heat transfer under varying welding conditions.

4.2.1. Trends in molten pool length

The simulation results in Fig. 11 reveal distinct trends in molten pool length across the experimental cases:

- **Cases 1 and 2:** Reducing the welding speed from 38 mm/s (case 1) to 13.8 mm/s (case 2) increased the molten pool length from 6.03 mm to 7.46 mm, a ~ 23.7 % increase. This is consistent with the increased

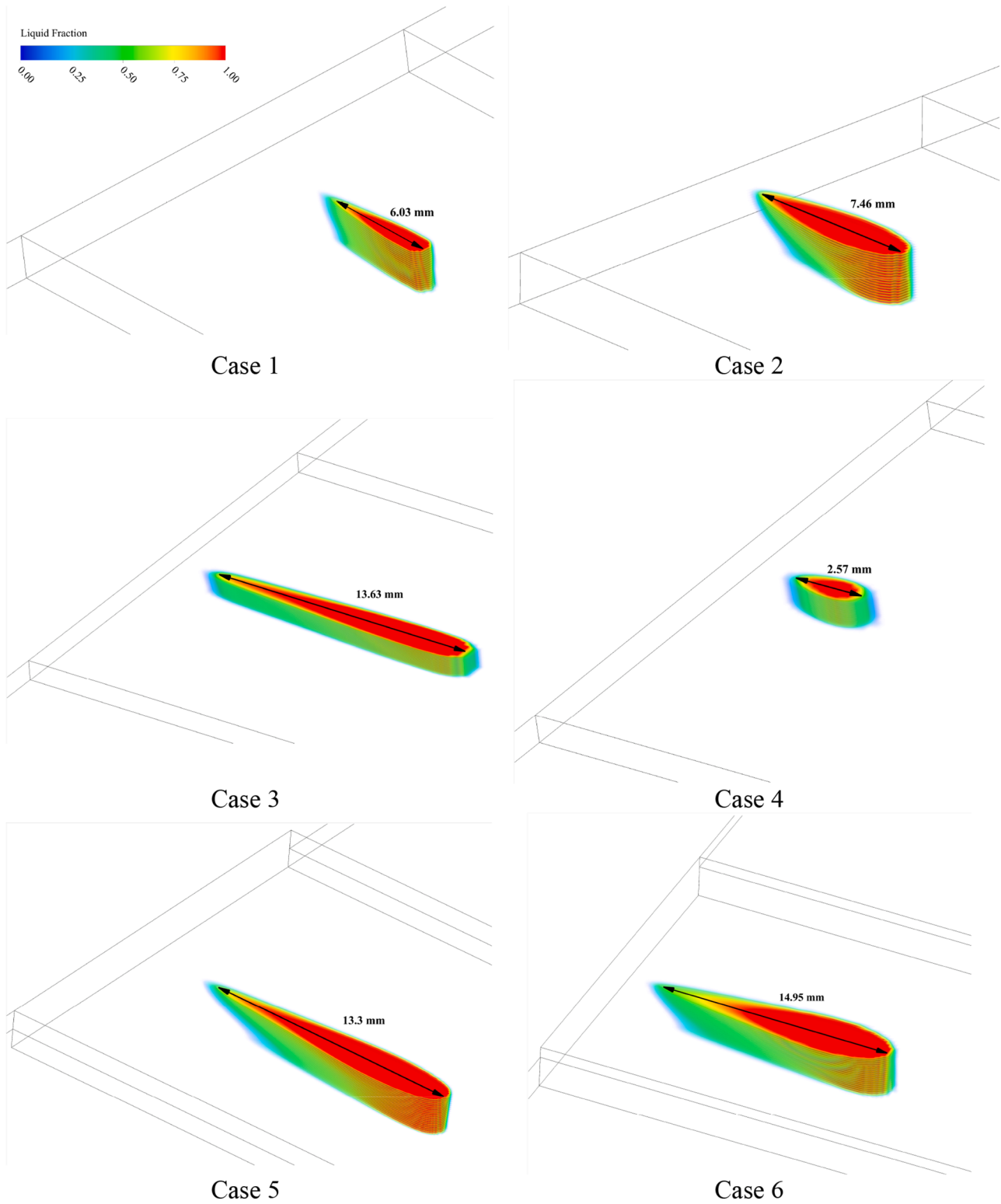


Fig. 11. Simulated molten pool lengths for all cases.

heat input per unit length at lower speeds, which prolongs the time the material remains molten, thereby elongating the pool. The increase in molten pool volume, further supports this trend, indicating a larger phase-affected region.

- **Cases 3 and 4:** Reducing the welding speed from 40 mm/s (case 3) to 14 mm/s (case 4) decreased the molten pool length from 13.63 mm to 2.67 mm, an $\sim 80.4\%$ reduction. This counterintuitive trend is

attributed to the head distance of 170 mm, which exceeds the focal distance (160 mm). At lower speeds, the defocused laser beam reduces penetration efficiency, leading to a smaller molten pool volume and a shorter pool length. This is consistent with the observed reduction in molten pool width in case 4, as the defocused beam concentrates energy less effectively.

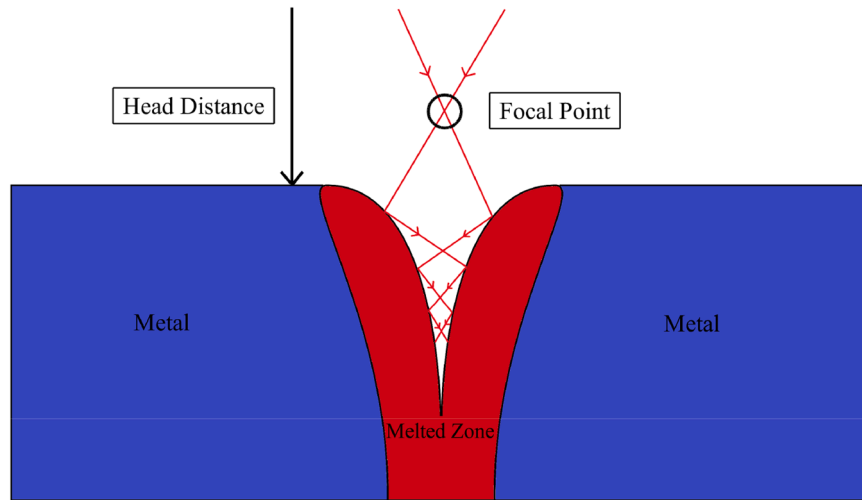


Fig. 12. Illustration of laser beam multi-reflection within the keyhole region when head distance exceeds focal length.

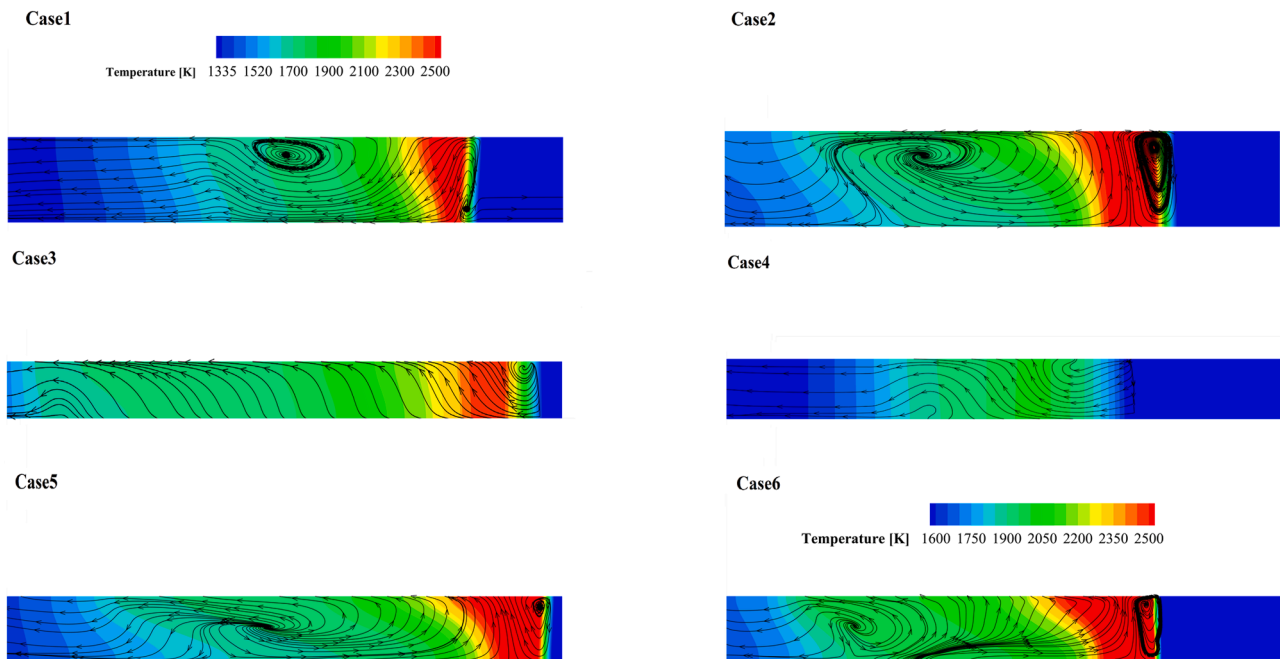


Fig. 13. Temperature distributions and streamline patterns in all cases, highlighting the influence of power, speed, and material properties.

- **Cases 5 and 6:** The molten pool lengths in cases 5 and 6 were 13.3 mm (316 L) and 14.95 mm (Ti6Al4V), respectively, significantly larger than in cases 1–4. These cases positioned the focal point on or within the workpiece, maximizing energy absorption and resulting in a voluminous molten pool. The longer pool length in case 6 compared to case 5 (~12.4 % increase) is attributed to the lower thermal conductivity and diffusivity of Ti6Al4V, which promote localized heat accumulation and slower cooling, thereby extending the molten zone longitudinally. The increased molten pool volume further supports the critical role of optimal beam focus in shaping weld pool dynamics.

4.2.2. Implications of molten pool length

The molten pool length significantly influences weld quality and process stability. Longer molten pools, as observed in cases 5 (13.3 mm) and 6 (14.95 mm), are associated with higher heat input and slower cooling rates, which may increase susceptibility to defects such as porosity and solidification cracks due to prolonged exposure to high

temperatures. Conversely, the short molten pool in case 4 (2.67 mm) indicates rapid solidification, which may enhance microstructural properties but increase the risk of incomplete fusion if parameters are not optimized. The model's ability to predict molten pool length trends enables the evaluation of welding parameters to achieve a balance between process stability and weld integrity.

The significant reduction in molten pool length in case 4 highlights the critical influence of head distance on weld pool dynamics. When the head distance exceeds the focal distance, the defocused beam reduces penetration efficiency, leading to a smaller and shorter molten pool. This finding underscores the importance of precise beam focus control in deep penetration welding to achieve desired weld profiles.

4.3. Flow behavior and temperature distribution in the molten pool

To further investigate the molten pool dynamics under varying process conditions, streamline and temperature contour plots were generated for six experimental cases involving both stainless steel 316 L

and Ti6Al4V. The results, visualized in Fig. 13, reveal significant differences in the flow patterns, temperature gradients, and molten pool geometries across different welding parameters.

In Case 1, a moderate laser power of 1400 W and a relatively high scanning speed of 38 mm/s resulted in an elongated molten pool with a primary recirculation vortex centered in the upper half of the pool. This indicates limited penetration and convection due to short interaction time and reduced energy input per unit length. In contrast, Case 2, with identical power but a lower speed (13.8 mm/s), exhibited a more concentrated heat zone and a stronger, deeper vortex structure. The extended interaction time allowed for enhanced heat transfer and deeper penetration, consistent with the broader molten region observed experimentally.

Case 3 and Case 4 shared the same power (1950 W) but differed in scanning speeds (40 mm/s vs. 14 mm/s). In Case 3, the high speed and thinner material produced a shallow pool dominated by surface-level heat dispersion, with minimal internal recirculation. However, in Case 4, the reduced speed led to a reversal in the vortex orientation and deeper thermal penetration. Interestingly, unlike the trend observed in Cases 1 and 2, the molten pool in Case 4 showed reduced length compared to Case 3. This deviation is attributed to the increased head distance (170 mm), which exceeded the focal length, diminishing the effective energy density at the workpiece and altering the beam's interaction with the material. This resulted in enhanced multi-reflection within the keyhole, thereby shifting the peak energy deposition to deeper regions.

In Case 5, a relatively balanced configuration (1900 W, 25 mm/s, 2 mm thickness) produced a stable molten pool with dual vortex structures and symmetric flow, indicative of effective melting. Finally, Case 6, the only case with Ti6Al4V, demonstrated unique flow behavior due to the alloy's lower thermal conductivity. A strong vortex formed near the rear end of the pool, accompanied by steep thermal gradients. The titanium sample's higher thickness (2.7 mm) and moderate speed resulted in a wide but deeper molten region, as expected from localized heat accumulation and reduced lateral dissipation.

Overall, the streamline analyses highlight the sensitivity of flow behavior to welding speed, power, material properties, and especially head distance relative to the focal point. Vortex formation, flow recirculation, and temperature gradients collectively influence the final shape of the molten pool. These results validate the ability of the numerical model to not only predict molten pool dimensions but also capture internal fluid dynamics critical to defect formation and metallurgical outcomes.

5. Conclusion

This study developed and validated a simplified numerical model for laser welding that balances predictive accuracy with low computational cost by combining a truncated Gaussian heat source and an enthalpy-porosity phase-change formulation. Validation against seven experiments (Ti6Al4V and 316 L) shows perfect agreement in molten-pool depth and absolute errors in top and bottom widths below 0.2 mm, while the model correctly reproduces trends in molten-pool length and internal flow patterns across a wide range of powers, speeds, head distances, materials, and thicknesses. The model achieves these results despite simplifying assumptions (neglect of vaporization/keyhole physics), demonstrating that calibrated reduced-order approaches can deliver industrially useful accuracy. Its low computational demand makes it practical for rapid parameter studies and preliminary process optimization, and the computed temperature fields provide a reliable input for subsequent residual-stress and distortion modeling. Overall, the proposed method offers a practical alternative to costly multiphysics simulations and a solid platform for future investigations into defect formation and thermo-mechanical response.

Statements and declarations

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

CRediT authorship contribution statement

Mahdi Ghamari: Writing – original draft, Visualization, Validation, Software. **Pouria Oliazadeh:** Writing – review & editing, Supervision, Project administration. **Mohammad Hossein Farshidianfar:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] D.H. Phillips, *Welding Engineering An introduction*, First, John Wiley & Sons, Southern Gate, 2016.
- [2] P. Li, Y. Fan, C. Zhang, Z. Zhu, W. Tian, A. Liu, Research on heat source model and weld profile for Fiber laser welding of A304 stainless steel thin sheet, *Adv. Mater. Sci. Eng.* (2018) 12, <https://doi.org/10.1155/2018/5895027>.
- [3] K. Oussaid, N. Omid, A. El Ouafi, N. Barka, Prediction of weld geometry in laser overlap welding of low-carbon galvanized steel, *Metals (Basel)* (2025) 15, <https://doi.org/10.3390/met15040447>.
- [4] D. Rosenthal, The theory of moving sources of heat and its application to metal treatments, *J. Fluids Eng. Trans. ASME* 68 (1946) 865, <https://doi.org/10.1115/1.4018625>.
- [5] P. Pavelic, V. R. Tanbakuhi, O.A. Uyehara, Myers, Experimental and computed temperature histories in gas tungsten arc welding of thin plates, *Weld. J.* 48 (1969) 295s–305s.
- [6] T. Kik, Heat source models in numerical simulations of laser welding, *Materials (Basel)* 13 (2020) 1–24, <https://doi.org/10.3390/ma13112653>.
- [7] T.W. Eagar, N.S. Tsai, Temperature fields produced by traveling distributed heat sources, *Weld. J.* 62 (1983) 346–355.
- [8] Z. Gao, P. Jiang, G. Mi, L. Cao, W. Liu, Investigation on the weld bead profile transformation with the keyhole and molten pool dynamic behavior simulation in high power laser welding, *Int. J. Heat Mass Transf.* 116 (2018) 1304–1313, <https://doi.org/10.1016/j.ijheatmasstransfer.2017.09.122>.
- [9] P. Xia, C. Wang, G. Mi, M. Zhang, L. Xiong, X. Zhang, C. Zhai, X. Feng, Y. Hu, Numerical simulation of molten pool flow behavior and keyhole evolution behavior in dual-laser beam oscillating bilateral synchronous welding of T-joints, *Int. J. Heat Mass Transf.* 209 (2023) 124114, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124114>.
- [10] Y. Ai, P. Jiang, C. Wang, G. Mi, S. Geng, Experimental and numerical analysis of molten pool and keyhole profile during high-power deep-penetration laser welding, *Int. J. Heat Mass Transf.* 126 (2018) 779–789, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.05.031>.
- [11] Y. Ai, P. Jiang, X. Shao, P. Li, C. Wang, G. Mi, S. Geng, Y. Liu, W. Liu, The prediction of the whole weld in fiber laser keyhole welding based on numerical simulation, *Appl. Therm. Eng.* 113 (2017) 980–993, <https://doi.org/10.1016/j.applthermaleng.2016.11.050>.
- [12] R. Lin, H. ping Wang, F. Lu, J. Solomon, B.E. Carlson, Numerical study of keyhole dynamics and keyhole-induced porosity formation in remote laser welding of Al alloys, *Int. J. Heat Mass Transf.* 108 (2017) 244–256, <https://doi.org/10.1016/j.ijheatmasstransfer.2016.12.019>.
- [13] F. Lu, X. Li, Z. Li, X. Tang, H. Cui, Formation and influence mechanism of keyhole-induced porosity in deep-penetration laser welding based on 3D transient modeling, *Int. J. Heat Mass Transf.* 90 (2015) 1143–1152, <https://doi.org/10.1016/j.ijheatmasstransfer.2015.07.041>.
- [14] N. Yadaiah, S. Bag, Development of egg-configuration heat source model in numerical simulation of autogenous fusion welding process, *Int. J. Therm. Sci.* 86 (2014) 125–138, <https://doi.org/10.1016/j.ijthermalsci.2014.06.032>.
- [15] M. Courtois, M. Carin, P. Le Masson, S. Gaied, M. Balabane, A new approach to compute multi-reflections of laser beam in a keyhole for heat transfer and fluid flow modelling in laser welding, *J. Phys. D: Appl. Phys.* 46 (2013), <https://doi.org/10.1088/0022-3727/46/50/505305>.
- [16] J. Li, P. Jiang, S. Geng, J. Xiong, Numerical and experimental study on keyhole dynamics and pore formation mechanisms during adjustable-ring-mode laser welding of medium-thick aluminum alloy, *Int. J. Heat Mass Transf.* 214 (2023), <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124443>.

- [17] P. Stavropoulos, G. Pastras, K. Tzimanis, T. Bekiaris, T. Souflas, An approach to modelling defocusing and keyhole reflectivity in keyhole laser processes, *Int. J. Adv. Manuf. Technol.* 134 (2024) 949–968, <https://doi.org/10.1007/s00170-024-14133-2>.
- [18] R... Rodger, E.D. Fletcher, The determination of internal stresses from the temperature history of a butt welded pipe, *Weld. J. Res. Suppl.* 17 (1938) 4–7.
- [19] K. MASUBUCHI, Control of distortion and shrinkage in welding, *Weld. Res. Council Bull.* 149 (1970) 1–30.
- [20] F.W. Brust, E.F. Rybicki, A computational model of backlay welding for controlling residual stresses in welded pipes, *J. Press. Vessel Technol. Trans. ASME*. 103 (1981) 226–232, <https://doi.org/10.1115/1.3263395>.
- [21] T.E. Barber, F.W. Brust, H.W. Mishler, M.F. Kanninen, Controlling Residual Stresses By Heat Sink Welding, Columbus, Ohio, 1981.
- [22] I. Sattari-Far, Y. Javadi, Influence of welding sequence on welding distortions in pipes, *Int. J. Press. Vessel. Pip.* 85 (2008) 265–274, <https://doi.org/10.1016/j.ijpvp.2007.07.003>.
- [23] M. Zain-ul-Abdein, D. Nelias, J.F. Jullien, D. Deloison, Prediction of laser beam welding-induced distortions and residual stresses by numerical simulation for aeronautic application, *J. Mater. Process. Technol.* 209 (2009) 2907–2917, <https://doi.org/10.1016/j.jmatprotec.2008.06.051>.
- [24] J.W. Kim, B.S. Jang, S.W. Kang, A study on an efficient prediction of welding deformation for T-joint laser welding of sandwich panel part II: proposal of a method to use shell element model, *Int. J. Nav. Archit. Ocean Eng.* 6 (2014) 245–256, <https://doi.org/10.2478/IJNAOE-2013-0176>.
- [25] J. Rahman Chukkan, M. Vasudevan, S. Muthukumaran, R.Ravi Kumar, N. Chandrasekhar, Simulation of laser butt welding of AISI 316L stainless steel sheet using various heat sources and experimental validation, *J. Mater. Process. Technol.* 219 (2015) 48–59, <https://doi.org/10.1016/j.jmatprotec.2014.12.008>.
- [26] G. Casalino, N. Contuzzi, F.M.C. Minutolo, M. Mortello, Finite element model for laser welding of titanium. *Procedia CIRP*, Elsevier B.V, 2015, pp. 434–439, <https://doi.org/10.1016/j.procir.2015.06.099>.
- [27] Y. Feng, X. Gao, Y. Zhang, C. Peng, X. Gui, Y. Sun, X. Xiao, Simulation and experiment for dynamics of laser welding keyhole and molten pool at different penetration status, *Int. J. Adv. Manuf. Technol.* 112 (2021) 2301–2312, <https://doi.org/10.1007/s00170-020-06489-y>.
- [28] J. Zhou, H.L. Tsai, P.C. Wang, Transport phenomena and keyhole dynamics during pulsed laser welding, *J. Heat Transfer*. 128 (2006) 680–690, <https://doi.org/10.1115/1.2194043>.
- [29] S. Pang, W. Chen, W. Wang, A quantitative model of keyhole instability induced porosity in laser welding of titanium alloy, *Metall. Mater. Trans. A Phys. Metall. Mater. Sci.* 45 (2014) 2808–2818, <https://doi.org/10.1007/s11661-014-2231-3>.
- [30] J.H. Cho, S.J. Na, Three-dimensional analysis of molten pool in GMA-laser hybrid welding, *Weld. J.* 88 (2009) 35–44.
- [31] H. Zhao, W. Niu, B. Zhang, Y. Lei, M. Kodama, T. Ishide, Modelling of keyhole dynamics and porosity formation considering the adaptive keyhole shape and three-phase coupling during deep-penetration laser welding, *J. Phys. D. Appl. Phys.* 44 (2011), <https://doi.org/10.1088/0022-3727/44/48/485302>.
- [32] J. Coleman, G.L. Knapp, B. Stump, M. Rolchigo, K. Kincaid, A. Plotkowski, A dynamic volumetric heat source model for laser additive manufacturing, *Addit. Manuf.* 95 (2024), <https://doi.org/10.1016/j.addma.2024.104531>.
- [33] F. Giudice, A. Sili, Validation of a theoretical model for laser welding thermal field by multi-physics numerical simulation, *Metals (Basel)* 13 (2023) 1–25, <https://doi.org/10.3390/met13122020>.
- [34] M. Courtois, M. Carin, P. Le Masson, S. Gaied, M. Balabane, Complete heat and fluid flow modeling of keyhole formation and collapse during spot laser welding, in: *ICALEO 2013 - 32nd Int. Congr. Appl. Lasers Electro-Optics* (2013) 77–84, <https://doi.org/10.2351/1.5062966>.
- [35] M. Jiménez-Xamán, M. Hernández-Hernández, R. Tariq, S. Landa-Damas, M. Rodríguez-Vázquez, A. Aranda-Arizmendi, P. Cruz-Alcantar, Numerical simulations and mathematical models in laser welding: a review based on physics and heat source models, *Front. Mech. Eng.* 10 (2024), <https://doi.org/10.3389/fmech.2024.1325623>.
- [36] A.B.H. Mekky, Computational modelling for specific heat and thermal conductivity of austenitic stainless steels alloys at solid phase, *Rev. Des Compos. Des Mater. Av.* 30 (2020) 23–27, <https://doi.org/10.18280/rcma.300104>.
- [37] Z. Guo, N. Saunders, J.P. Schillé, A.P. Miodownik, Material properties for process simulation, *Mater. Sci. Eng. A*. 499 (2009) 7–13, <https://doi.org/10.1016/j.msea.2007.09.097>.
- [38] P. Hartwig, N. Bakir, L. Scheunemann, A. Gumenyuk, J. Schröder, M. Rethmeier, A physically motivated heat source model for laser beam welding, *Metals (Basel)* 14 (2024) 1–26, <https://doi.org/10.3390/met14040430>.
- [39] C.Y. Liang, S.N. Lan, Y.S. Wang, Review: High-Performance Dissimilar Metal Joining Between Titanium Alloys and Steel, Springer US, 2025, <https://doi.org/10.1007/s10853-025-11277-8>.
- [40] Z. Guo, N. Saunders, A. Miodownik, J.P. Schillé, Modelling properties of titanium alloys : during solidification or at elevated temperatures, *Rare Met. Mater. Eng.* 29 (2006) 2, <http://www.cntech.com.cn/online/h000/h12/attach200704131624130.pdf>.
- [41] V.V. R. P. C. A fixed grid numerical modelling methodology for convection diffusion mushy region phase change problems, *Int. J. Heat Mass Transf.* 30 (1978) 1709–1719.
- [42] A. Bejan, *Convection Heat Transfer*, forth edit, Wiley & Sons, 2013, pp. 53–101, https://doi.org/10.1007/978-1-4939-2565-0_3.
- [43] M. Dal, R. Fabbro, An overview of the state of art in laser welding simulation, *Opt. Laser Technol.* 78 (2016) 2–14, <https://doi.org/10.1016/j.optlastec.2015.09.015>.
- [44] M. Behúlová, E. Babalová, Heat source models for numerical simulation of laser welding processes - A short review, *J. Phys. Conf. Ser.* (2024) 2712, <https://doi.org/10.1088/1742-6596/2712/1/012018>.